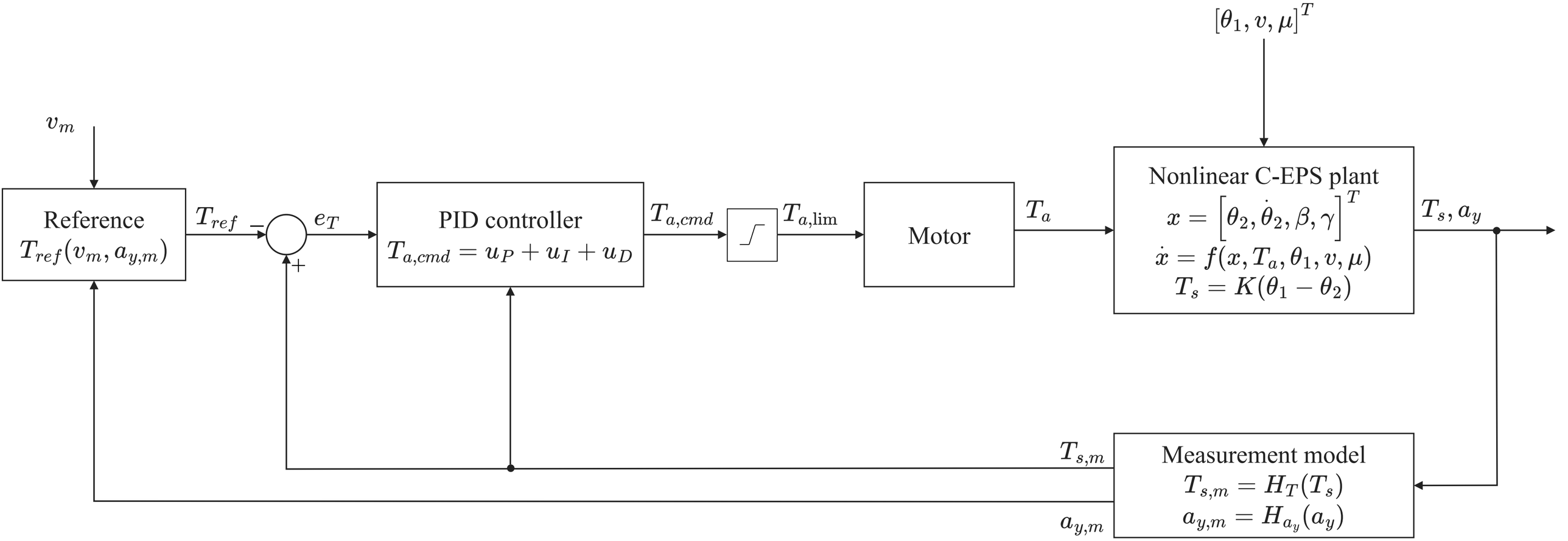
$[\theta_1, v, \mu]^T$
v_m
Reference
$T_{ref}(v_m, a_{y,m})$
T_{ref}
$-$
$+$
e_T
PID controller
$T_{a,cmd} = u_P + u_I + u_D$
$T_{a,cmd}$
$T_{a,\text{lim}}$
Motor
T_a
Nonlinear C-EPS plant
$x = \left[\theta_2, \dot{\theta}_2, \beta, \gamma\right]^T$
$\dot{x} = f(x, T_a, \theta_1, v, \mu)$
$T_s = K(\theta_1 - \theta_2)$
T_s, a_y
$T_{s,m}$
$a_{y,m}$
Measurement model
$T_{s,m} = H_T(T_s)$
$a_{y,m} = H_{a_y}(a_y)$